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## Third Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026 Object Oriented Programming with Java

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                              | M | L  | C   |
|------------|----|----------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | Explain feature of Java.                                                                     | 7 | L2 | CO1 |
|            | b. | Define array, write a java program to calculate the average among the elements [8, 6, 2, 7]. | 7 | L3 | CO1 |
|            | c. | List and explain operators in JAVA with examples.                                            | 6 | L2 | CO1 |
| OR         |    |                                                                                              |   |    |     |
| Q.2        | a. | Explain OOP's features in java.                                                              | 7 | L2 | CO1 |
|            | b. | Write a java program to sort the elements using a for loop.                                  | 7 | L3 | CO1 |
|            | c. | With example, explain different types of if statement in JAVA.                               | 6 | L2 | CO1 |
| Module – 2 |    |                                                                                              |   |    |     |
| Q.3        | a. | Define Constructor. Explain two types of constructors with an example.                       | 7 | L3 | CO2 |
|            | b. | Define Recursion. Write a recursive program to find factorial of a number.                   | 7 | L3 | CO2 |
|            | c. | Explain garbage collection with an example, explain final and finalize ( ) method.           | 6 | L2 | CO2 |
| OR         |    |                                                                                              |   |    |     |
| Q.4        | a. | Define class. Explain call by value and call by reference with an example program.           | 7 | L3 | CO2 |
|            | b. | Using proper class and methods write a program to perform stack operations.                  | 7 | L3 | CO2 |
|            | c. | Explain the use of this keyword in java with an example.                                     | 6 | L2 | CO2 |
| Module – 3 |    |                                                                                              |   |    |     |
| Q.5        | a. | Write a java program to implement multilevel inheritance with 3 levels of hierarchy.         | 7 | L3 | CO3 |
|            | b. | Define interface. With suitable program explain nested interface in java.                    | 7 | L3 | CO3 |
|            | c. | Explain dynamic method dispatch with a suitable example.                                     | 6 | L2 | CO3 |
| 1 of 2     |    |                                                                                              |   |    |     |

## OR

|     |    |                                                                                                  |   |    |     |
|-----|----|--------------------------------------------------------------------------------------------------|---|----|-----|
| Q.6 | a. | Explain inheritance. Write a java program to implement single level inheritance.                 | 7 | L3 | CO3 |
|     | b. | Explain the importance of the super key word in inheritance, illustrate with a suitable example. | 7 | L3 | CO3 |
|     | c. | Define method overloading and overriding with example.                                           | 6 | L2 | CO3 |

## Module – 4

|     |    |                                                                                                     |   |    |     |
|-----|----|-----------------------------------------------------------------------------------------------------|---|----|-----|
| Q.7 | a. | Define Package, with an example, explain the steps are involved in creating a user-defined package. | 7 | L2 | CO4 |
|     | b. | With sample code, explain chained exception.                                                        | 7 | L3 | CO4 |
|     | c. | Define an exception, with syntax explain all five keywords used in exception handling.              | 6 | L2 | CO4 |

## OR

|     |    |                                                                             |   |    |     |
|-----|----|-----------------------------------------------------------------------------|---|----|-----|
| Q.8 | a. | Explain the concept of package importing in java with an example.           | 7 | L2 | CO4 |
|     | b. | How do you create your own exception class, explain with a program.         | 7 | L2 | CO4 |
|     | c. | With an example, explain working of a nested try block within an exception. | 6 | L3 | CO4 |

## Module – 5

|     |    |                                                                                      |   |    |     |
|-----|----|--------------------------------------------------------------------------------------|---|----|-----|
| Q.9 | a. | Define Thread. With diagram explain the java thread model.                           | 7 | L2 | CO5 |
|     | b. | Explain synchronization with an example, how synchronization is implemented in java. | 7 | L3 | CO5 |
|     | c. | With suitable example, explain values( ) and valueOf( ) method in enumeration.       | 6 | L2 | CO5 |

## OR

|      |    |                                                                                         |   |    |     |
|------|----|-----------------------------------------------------------------------------------------|---|----|-----|
| Q.10 | a. | Define Multithreading, write a program to create multiple threads in java.              | 7 | L2 | CO5 |
|      | b. | Demonstrates the usage of compareTo( ) and equals( ) method with enumeration constants. | 7 | L3 | CO5 |
|      | c. | Explain autoboxing / unboxing in expressions.                                           | 6 | L2 | CO5 |

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